

Respected teacher and dear fellow classmates, a very good noon to you all. Today, I am going to present a topic that is changing the world of technology – Neural Networks.

Okay, let's start with a simple question: Have you ever wondered how your smartphone can recognize your face or how Netflix recommends your favorite show? Let's see how this amazing technology works!

A neural network is inspired by how our brain works. Just like our brain has billions of neurons communicating with each other, a neural network has artificial neurons that process information. These networks learn patterns from examples, so they can make decisions or predictions on new data.

Neural networks work in layers. The input layer takes information, hidden layers process it, and the output layer gives the result.

Input Layer: The input layer contains 3 nodes that indicates the presence of each keyword.

Hidden Layer: The input vector is passed through the hidden layer. Each neuron in the hidden layer performs two primary operations: a weighted sum followed by an activation function.

Output Layer: The activated values from the hidden neurons are sent to the output neuron where they are again processed using a weighted sum and an activation function.

Neural networks are everywhere! They power AI assistants like Siri and Alexa, help doctors detect diseases, predict stock market trends, and even make video games smarter. They are truly revolutionizing technology in every field.

To conclude my part, neural networks are a fascinating technology that teaches machines to think, learn, and make decisions, inspired by our brain. Thank you everyone for patiently listening.

Now, I would like to invite our next speaker to continue the presentation.